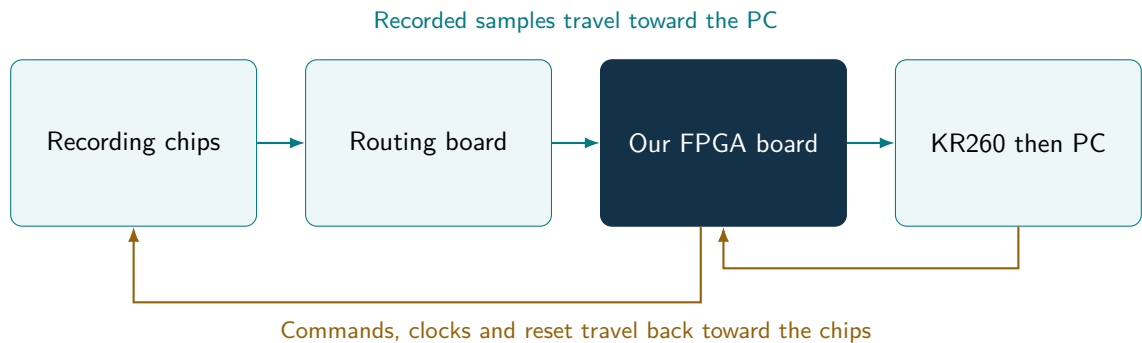


START HERE / WHAT THE REFERENCES TEACH US

Understanding our FPGA board

The meeting settled the job; it did not finish the interfaces. Build a compact Artix-7 board that receives the eight-chip recording system, starts automatically, and exchanges data and commands with KR260. This guide answers the six questions from the September 17 notes with evidence and explicit remaining decisions. [1]



Three references, three different lessons

Reference	What to learn from it
CP SOM ONE old open-source PCB	Artix-7 support: FPGA, power, capacitors, oscillator, boot flash and headers. It does not already implement our eight-chip recorder or KR260 link. [2]
PCB-5 new chip-board example	Two custom recording chips and an 80-contact connector. Its pin worksheet helps define the eight-chip interface; its chip revision still needs Gerald’s confirmation. [3]
LDO board older routing example	Nine recording-chip regulators and two 50-contact connectors. Shows power and passive signal routing; it does not combine data in logic. [4]

A first walk through the open-source project in KiCad

Start with CP_SOM_ONE_Working_Copy/kicad/som_fpga.kicad_pro inside 9_16_26. The **schematic** shows electrical connections; the **PCB** shows the physical arrangement and copper. In the schematic, follow these sheets:

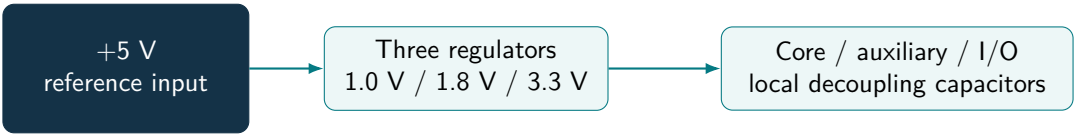
- **supply** (sheet 7): +5 V becomes the reference’s three FPGA supply rails.
- **FPGA_bank0_vss** (5): configuration/startup pins and FPGA power/ground.
- **FPGA_bank13_14** (3, flash), **FPGA_bank15_16** (4, clock), **FPGA_bank34_35** (2): I/O groups called *banks*; each bank’s VCCO supply constrains its signaling voltage.
- **connectors** (8): where signals leave the module. **FPGA_mgt** (6) shows unused transceiver circuitry; this board does not provide a ready high-speed serial link.

Reading key: **Known** means found in files or agreed in the meeting, not bench-tested. **Proposed** means a candidate implementation. **TBD + owner** identifies a decision still required. Original hardware files were not redesigned for this guide.

QUESTION 1 / PART + MINIMUM SUPPORT

What must surround the FPGA?

Answer: use XC7A35T as the meeting baseline, then confirm its full ordering code. A working board still needs correct power, local capacitors, configuration/startup settings, a usable clock plan, and application/debug connections. “Minimum” is a set of functions, not a safe universal component count. [1,2,6]



This diagram describes CP SOM ONE. The new board’s I/O rail must match the actual chip/KR260 interfaces; 3.3 V is not an approved replacement for a 1.5 V interface.

Function	Known in CP SOM ONE	What our board needs
FPGA	U1 says XC7A35T-FGG484 in schematic and PCB.	Confirm density, package, speed and temperature grade before purchase.
Power	U2–U4 generate 3.3 V, 1.8 V and 1.0 V; rail capacitors surround the FPGA.	Current estimate, voltage/bank plan, stable regulation, ramp/sequence and layout review.
Configuration	U5: MX25L12833F, 128 Mbit flash; mode and configuration resistors.	Retain a reviewed boot path plus initial programming and recovery (question 5).
Clock	X1: SG-8018CE; source article says 30 MHz.	Resolve actual frequency/order code and derive/accept the needed ASIC and link clocks.
Connections	J2–J4 expose signals, power and JTAG; J1 selects mode.	New connector/pin plan, timing constraints, ground returns and test access.
Optional indicators	Power, DONE and user LEDs are present.	Choose useful diagnostics; removing an LED does not remove required startup bias.

Three conflicts to resolve before copying the circuit

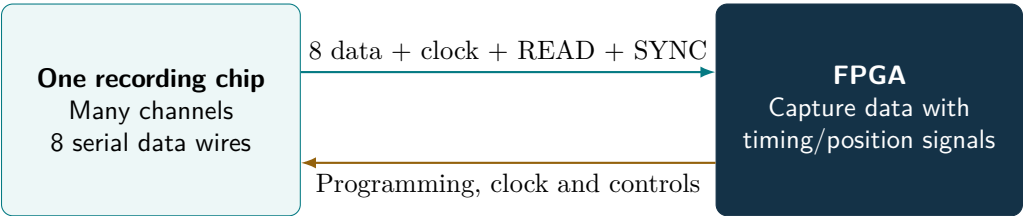
- 35T versus 75T.** The older production BOM lists XC7A75T-FGG484; current schematic/PCB use 35T. The BOM’s supplier code previously mapped to a 75T part. Do not silently turn that into the approved 35T purchase. [2]
- Regulator identity.** U2–U4 value fields say TLV62565, while their symbol/datasheet metadata identifies NCP1529A; U3/U4 supplier metadata adds to the conflict. Reconcile actual parts before applying a stability/current specification. [2]
- C44–C46 are DNP in the current schematic.** They have feedback-to-ground connectivity, but *do not populate* flags contradict treating the old BOM/audit as a confirmed assembled circuit. Check the selected revision and assembly intent. [2]

TBD + owner: Howard and Zitong, with Gerald confirming the baseline. Produce one reconciled schematic/BOM and a power/clock/bank table. The reference’s saved design is evidence of intent, not electrical validation.

QUESTION 2 / SIGNALS + DIRECTIONS

What does “88 signals” include?

Answer: 88 plausibly counts chip outputs only. Gerald’s worksheet lists eight data wires plus returned clock, READ and SYNC for each chip. Eight chips therefore produce $8 \times (8+1+1+1) = 88$ signals entering the FPGA, before programming, other controls, power and ground. [1,3]



Worksheet group	Eight chips	Direction	Status
8 data wires per chip	64	Chip → FPGA	Listed chip outputs.
CLK32MHZ, READ, SYNC	24	Chip → FPGA	Listed chip outputs; timing still needs validation.
SPI_DL, SPI_DR	16	FPGA → chip	Recording-chip setup inputs, separate from FPGA boot.
Shared control/test entries	11	Resolve per signal	Sharing is a planning assumption, not approved fan-out.
DISC_OUT	0 or 8	Chip output candidate	Worksheet leaves use open.
Power, references, grounds	Additional	Separate power plan	Do not count these as spare digital I/O.

Why the meeting estimate still needs reconciliation

Keeping all 11 shared entries gives the **conditional** total $88+16+11 = 115$ signal nets, or 123 with eight DISC_OUT signals. These are worksheet arithmetic, not approved requirements. The 11 worksheet entries are CLKH, CHIP_RST, FE_RESET, SPI_CLK, SPI_LATCH, STIM_START, STM_CLK, STIM_EN, STIM_CHB, AC_IN and IMP_TST. Names differ slightly in other files. AC_IN/IMP_TST are grouped as shared controls here, but legacy HDL treats their counterparts as FPGA spare inputs; the new schematic uses passive pin types. Gerald must resolve their directions. [3,5]

CP SOM ONE exposes 76 FPGA-connected header signals plus four JTAG nets at its headers. Even 88 chip outputs exceed that exposed signal count, before the downstream link. The FPGA package and the reference module’s connector breakout are different limits: retaining the chip does not require retaining those headers. A fresh bank/pin plan is needed. [2]

Voltage check before wiring: the old FPGA_512 constraints specify 1.5 V ASIC signaling; CP SOM ONE’s exposed I/O banks use 3.3 V. Verify the current chip revision, then design compatible banks or justified translation. Do not connect them based on matching signal names. [2,5,6]

TBD + owner: Gerald supplies the chip truth; Howard and Zitong reconcile the board interfaces. For every net record: chip/revision, direction, shared/per-chip, voltage, clock edge/rate, inactive state, FPGA bank/pin and connector contact. [1]

QUESTION 3 / DATA RATE + LANE COUNT

How fast must the KR260 link be?

Answer: calculate the sustained byte rate before choosing two or four lanes. Under the old sample/framing format, eight 512-channel chips require **132 or 264 MB/s**, depending on sample rate. The sample-rate conflict is still open; these are sizing scenarios, not measured performance. [1,5]

A transparent calculation

Assume 4,096 channels, 12-bit ADC values, no compression, and one 16-bit transferred word per sample. The legacy frame carries 128 data words plus four framing words:

$$R_{\text{framed}} = 4096 \times f_s \times 2 \frac{\text{bytes}}{\text{sample}} \times \frac{132}{128}.$$

Per-channel rate	15,625 samples/s	31,250 samples/s
Where it appears	Old host software	Gerald’s earlier slides
Packed 12-bit values	96 MB/s	192 MB/s
16-bit sample words	128 MB/s	256 MB/s
With old framing	132 MB/s	264 MB/s
Corresponding bits/s	1.056 Gbit/s	2.112 Gbit/s

MB/s uses decimal millions of bytes. Framing is the legacy 132/128 factor only; add the chosen link’s coding/protocol overhead, controls and operating margin. The new format may differ. [5]

What “two to four data lines” would require

At the higher scenario, dividing 2.112 Gbit/s evenly gives **1.056 Gbit/s per lane for two lanes**, or **528 Mbit/s per lane for four lanes**, before additional overhead. These are minimum arithmetic rates; they do not prove Artix I/O timing, signal integrity, cable operation or a working KR260 receiver.

Also define what a *line* means. A differential lane needs two signal conductors; a source-synchronous interface may need clock and control conductors as well. Ground/return contacts, reset and commands still occupy connector space.

A pause creates a memory requirement

If the receiver cannot accept data for a time t , the FPGA needs available buffering of at least $R \times t$, plus margin. At 264 MB/s, a 10 ms complete stall accumulates **2.64 MB**. Boot flash stores the configuration image when power is off; a recording buffer temporarily holds incoming samples. These are different jobs. The 35T has 1,800 Kbit total block RAM (225 KiB including parity capacity, before allocation/format limits). Thus 10 ms at this rate cannot fit even in all block RAM: shorter stalls, downstream flow control if feasible, or more buffering would be needed. [6]

TBD + owner: Gerald confirms the recording mode; David and the firmware collaborator agree the receiving path with Howard/Zitong. Specify channel count, sample rate/format, sustained rate, maximum stall, loss policy, controls and error reporting. Validate with sustained capture and overflow/sequence counters, not only a visible plot. [1,5]

QUESTION 4 / CONNECTOR + PROTOCOL

What should connect to KR260?

Answer: choose an implemented link at both ends, then choose its cable. USB-C and micro-HDMI describe possible connector choices from the meeting. No USB or HDMI protocol was agreed, and connector shape does not make an FPGA implement either protocol. [1,9]

KR260 is more capable than the meeting’s “MCU” shorthand

KR260 uses a K26 module with a Zynq UltraScale+ MPSoC. Its carrier has four USB 3 host ports, four 1 Gb/s Ethernet ports, Pmod/HAT expansion, SLVS-EC, SFP+ and a DisplayPort *output*. Each path still needs compatible hardware and receiver software. One 1 Gb/s Ethernet link has a raw ceiling of 125 MB/s, already below either *framed* scenario; USB 2 high speed is also insufficient. [8]

Candidate path	What would make it concrete	Trade-off to resolve
USB through a bridge	Add a suitable USB bridge/controller, FPGA-side interface and KR260 host driver/application. Verify sustained rate.	Familiar cable; bridge, power and software add area/work. A direct FPGA-to-USB socket is not enough.
Source-synchronous data into KR260 logic	Name accessible receiving pins, voltage, clock/data lanes and timing constraints; implement receiver and buffering.	Potentially simple custom logic; cable/clock integrity, pin access and voltage compatibility need proof.
High-speed serial link	Identify usable transceiver resources and physical routes on both selected boards; choose framing, clocking and receiver logic.	Fewer data lanes can require more specialized hardware and firmware. CP SOM ONE does not expose a useful implemented transceiver path; redesign is required.

These are architecture candidates, not confirmed KR260 implementations or endorsed component selections. The local CP SOM ONE board does not already supply one of these acquisition links.

Six items belong on the same interface drawing

1. **Electrical scheme:** voltage, single-ended/differential signaling and return paths.
2. **Data transport:** lane count, usable rate, framing and error detection.
3. **Commands:** start/stop, chip setup, resets, status and fault handling.
4. **Physical mapping:** exact connector pairs, cable conductors, pinout and orientation.
5. **Power behavior:** who supplies power; what happens when only one board is on.
6. **Implementation ownership:** FPGA transmitter, KR260 receiver and application.

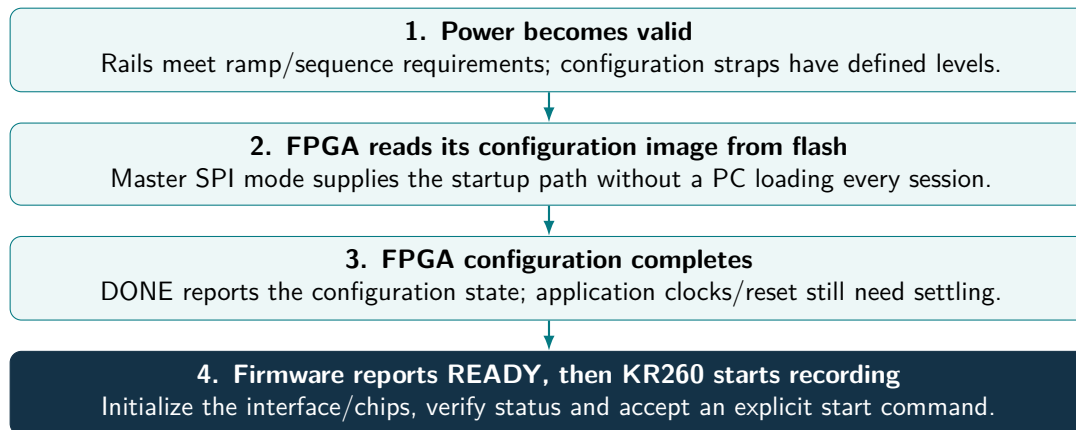
TBD + owner: Howard/Zitong + David + FPGA firmware collaborator. Compare one concrete implementation per candidate against question 3. If a familiar connector carries a custom pinout, explicitly prevent connection to an ordinary USB/HDMI device. The proposed downstream adapter board also needs an owner and schematic. [1]

Practical first deliverable: one page showing both endpoint parts, actual accessible pins, direction/voltage, cable mapping and a rate budget. Only then freeze the connector footprint.

QUESTION 5 / INITIAL PROGRAMMING + BOOT

How can users just turn it on?

Proposed answer: retain local configuration flash for automatic startup. Program its configuration image (the *bitstream*) during development/assembly, then let the FPGA load itself at power-on. CP SOM ONE contains the relevant flash and mode circuitry; this review has not demonstrated boot on a physical board. [2,7]



What the reference tells us

- **U5 is MX25L12833F, 128 Mbit = 16 MiB.** Its stored schematic datasheet link names a different flash family/density. AMD's current Artix-7 configuration-memory list includes MX25L12833F; still select the exact part, voltage, mode and programming settings. [2,7]
- **J1 open: M[2:0] = 001 (Master SPI); shunted: 101 (JTAG-only). JTAG remains accessible in Master SPI mode.** Confirm the physical assembly/straps before attempting either flow. J1 is marked DNP in the source. [2,6]
- **R8 is 4.7 kΩ from M2 to ground.** AMD's configuration guidance calls for mode-pin pull resistors no greater than 1 kΩ; review this source discrepancy. PUDC_B is exposed at J4 pin 26, so its configuration-time bias also needs definition. [2,6]

Initial programming and later updates are separate procedures

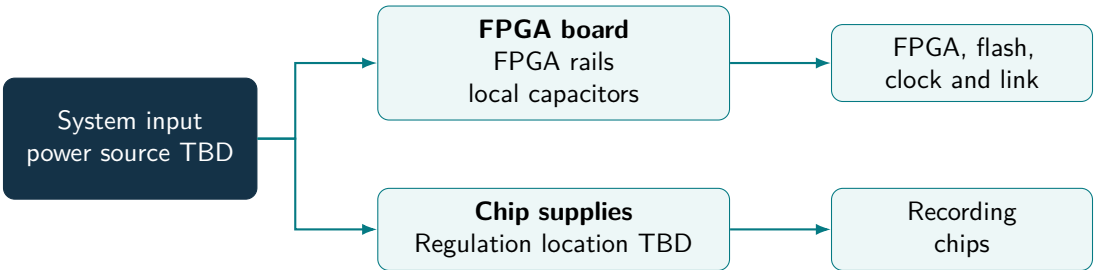
Keep accessible JTAG signals, ground and a correct voltage reference. An engineer can program the configuration memory through a suitable programming flow, verify it, select the intended boot mode and power-cycle to test autonomous startup. Define recovery after an interrupted update. A later KR260-driven updater is a separate hardware/firmware feature, not something this reference proves. [7]

TBD + owner: Howard/Zitong + FPGA firmware collaborator; David for KR260 involvement. Approve flash image/settings, mode/bias circuit, programmer access, initial programming instructions, READY/error behavior and recovery. **DONE does not mean the complete recording application is ready.** READY is proposed firmware status, not an existing reference-board pin.

QUESTION 6 / POWER PLACEMENT + SIZE

Where do supplies go, and how small?

Answer: FPGA power stays on the FPGA board, as agreed. Recording-chip regulator placement remains open. Set connector, signal/bank, power and assembly requirements before selecting final dimensions or copper-layer count. [1]



Why the nine LDOs are useful, but not a finished power plan

The older LDO board generates named recording-chip supply/reference nets. An LDO is a linear regulator; the LT3042 family is designed for low noise. Decide voltage, load current, ripple/noise, input headroom and heat for the selected eight-chip arrangement. Then decide placement, connectors and local decoupling. Do not assume nine reference regulators scale unchanged to eight chips. [4,10]

Some printed supply labels disagree with their connected nets: for example, U6 connects to VDD_CGEN while nearby printing says VDD_ADC. Its symbol/package/folder naming also needs reconciliation. Identify a rail by the schematic/net connection, not the closest printed label. [4]

Saved reference	Approx. outline	Copper	Useful context
CP SOM ONE	60 × 50 mm	4 layers	FGG484 package body: 23 × 23 mm.
PCB-5	22 × 17 mm	10 layers	Two chips, 80-contact connector.
LDO reference	18.3 × 42 mm	6 layers	Two 50-contact main connectors.

These are reference dimensions/layers, not proposed requirements for our FPGA board. PCB-5’s enclosed fabrication export differs from the saved PCB; select a revision before making new manufacturing files. [2–4,6]

A smaller board is a whole-assembly decision

Connector footprints, BGA escape routing, ground returns, decoupling, regulator heat, programming access and assembly rules all occupy space. The 80-contact chip connector and older 50-contact connectors are not a confirmed mating pair. The meeting’s two detachable board-to-board connectors still need exact pairs, orientation and stack height. [1,3,4]

TBD + owner: Gerald defines chip rails; Howard and Zitong allocate power and mechanics. Agree maximum width/length/height, current/thermal budget, connector placement, assembly limits and test access. Then choose a stack-up that can route the selected BGA/interface and obtain fabrication review. JLC preference does not settle the layer count. [1]

NEXT REVIEW / TURN THE ANSWERS INTO DECISIONS

Bring these six concrete results

Most useful next step: create one shared interface drawing and signal table. It connects the new chip-board examples to the old FPGA support design and gives the firmware collaborators something specific to evaluate. Keep uncertain fields labeled **TBD + owner**.

	Result to bring	What must be written down	Owner / status
1	Reconciled FPGA baseline	Full FPGA ordering code; regulator identities; C44–C46 DNP intent; clock part/frequency; voltage/current/bank table.	Howard + Zitong; Gerald confirms. TBD
2	Eight-chip signal table	Each retained net's direction, shared/per-chip status, voltage, timing, idle state and connector/FPGA pins. Resolve AC_IN, IMP_TEST, DISC_OUT.	Gerald supplies; Howard + Zitong reconcile. TBD
3	Recording-rate contract	Channels, rate, sample format, framing, sustained throughput, maximum pause, buffer/loss policy and error reporting.	Gerald + firmware collaborator + David. TBD
4	One link drawing	Both endpoints and accessible pins; protocol, data/clock/control lanes, electrical levels, exact cable and connectors.	Howard + Zitong + firmware collaborator + David. TBD
5	Startup/update procedure	Initial programming, automatic flash boot, mode/bias review, READY handshake, update verification and recovery.	Howard + Zitong + firmware collaborator. Proposed flash route TBD details
6	Power/mechanical envelope	Chip-regulator location and specifications; board outline/height; heat, connector placement, layer/assembly constraints.	Gerald + Howard + Zitong. TBD

What Howard can confidently say now

- “I understand the blocks: chip and routing references define the input; CP SOM ONE teaches FPGA support circuitry; the KR260 link still needs implementation choices.”
- “The 88 count excludes programming and other controls. I need an approved per-signal table before assigning FPGA pins.”
- “Local flash is a plausible way to meet power-on usability. I will keep configuration completion separate from application readiness.”
- “We can estimate the required link rate now, while keeping the recording mode and lane/protocol decisions open.”

What evidence should close the decisions?

A reconciled schematic/BOM and bank/pin plan close the paper design questions. Electrical rules/layout review, timing closure, programming/recovery tests and sustained end-to-end capture are separate checks needed before claiming a working board. None was established just by opening these references.

Scope reminder: high-pass/low-pass filtering was discussed as a possible future function, not an immediate agreed deliverable. Include it in resource planning only with confirmed requirements. [1]

SOURCE REGISTER / WHERE EACH ANSWER COMES FROM

Evidence and useful reading

Local paths below are relative to the FPGA PCB folder. The guide follows the September 17 meeting; older reference material informs implementation but does not overrule newer decisions. Sources were reviewed September 17, 2026.

- [1] **September 17 meeting notes.** 9_17_26/FPGA_PCB_Meeting_Notes_2026-09-17.docx (PDF alongside). Main outcome, agreed direction, six questions and owners. This is the scope authority for this guide.
- [2] **CP SOM ONE, actual working-copy source.** 9_16_26/CP_SOM_ONE_Working_Copy/kicad/. Checked schematic hierarchy, PCB, generated current netlist and older production BOM. Original snapshot: b272899d62ac2e454f85c413d64d62271137da27. [Upstream project](#); [author's design article](#). Older local audits in 9_16_26/research/ are useful context, but the current C44-C46 DNP flags take precedence over prior assembly assumptions.
- [3] **PCB-5 and pin worksheet.** 9_17_26/PCB-5/PCB.kicad_sch, PCB.kicad_pcb and Pins.xlsx, Sheet1. Chip/net structure, connector and conditional signal-count arithmetic. Does not establish final chip revision or eight-chip compatibility.
- [4] **LDO/routing example.** 9_17_26/LD0_Board_10S0IC (New Rigid)/, schematic and PCB. Power/routing topology, connectors and annotation conflicts. A fuller local explanation is 9_17_26/review/Understanding_Geralds_Two_Boards.md.
- [5] **Legacy recording interface.** 9_16_26/reference/FPGA_512/adc_4lane_framed/: top_4lane_framed.v, adc_4lane_framed.lpf, adc_live.py; Gerald's 9_16_26/research/gerald_slides/Chip_FPGA_Interface.pptx, especially slides 1, 3, 5, 14, 15, 25 and 27. The ECP5 reference supplies interface/format evidence; its bitstream and physical pins do not transfer to Artix-7.
- [6] **AMD FPGA electrical/configuration guidance.** [DS180: Table 4, block RAM](#); [DS181: power/electrical limits](#); [UG470: configuration modes and pin bias](#); [UG471: I/O banks](#); [UG483: PCB/decoupling](#); [UG475: package/pinout](#). Local copies: 9_16_26/Datasheets/AMD/. Apply the selected part's conditions, not an isolated typical value.
- [7] **AMD Vivado configuration-memory documentation.** [UG908: Artix-7 configuration memory devices; programming a configuration memory device](#). These support flash selection/programming workflow, not a claim that our board has booted.
- [8] **AMD KR260 documentation.** [DS988: product details](#); [UG1092: carrier interfaces](#). Check the actual board schematic and receiving implementation before assigning a custom cable.
- [9] **USB-IF connector terminology.** [USB Type-C language and packaging guidelines](#). Connector naming alone does not specify the complete data implementation.
- [10] **Analog Devices recording-supply reference.** [LT3042 product page and datasheet](#). Confirms regulator family; it does not settle the new board's rails, package selection or load budget.

Guide revision 1. Diagrams are original explanatory drawings. This document is a source-based learning and decision guide: no component removal, final pinout, fabrication release, programmed hardware, timing closure or successful recording is claimed. Editable source: Gerald_Meeting_Answers_Guide.tex.